

Students

Unique. + Not Null

↓

name	email	phone	---
101	abc@---	Null	---

→

Primary Key → Unique
→ Not Null.

↳ Uniquely identify a record.

↳ Email id column can be the PK.

→ Email id can be changed.

→ String Data type.

↳ Search can be slow.

→ Index table Size ↑

PK = System Generated.

PK { Unique + Not Null

Students

Unique + Not Null

id	name	email	phone	- - -
1		abc@-	Null	
2				
3				
4				
5				
6				
...				
1				

1. Masai will have multiple **batches**.
2. For each **batch**, we need to store the name, start month and current instructor.
3. Each **batch** of Masai will have multiple **students**.
4. Each batch has multiple **classes**.
5. For each **class**, store the name, date and time, instructor of the class.
6. For every **student**, we store their name, graduation year, University name, email, phone number.
7. Every student has a **buddy**, who is also a student.
8. A student may move from one **batch** to another.
9. For each batch a **student** moves to, the date of starting is stored.
10. Every student has a **mentor**.
11. For every **mentor**, we store their name and current company name.
12. Store information about all **mentor sessions** (time, duration, student, mentor, student rating, mentor rating).|
13. For every **batch**, store if it is an AIML batch or a SDE batch.

Students
id (PK)
name
email → Unique + NOT NULL
grad-year
phone
==

batches
batch-id (PK)
name
Start-month
—

Instructors
id (PK)
name
—
—
—

Classes
Class-id
title
—
—

mentors
—
—
—
—

mentor-sessions
—
—
—
—
—

Cardinality

→ how 2 entities are related to each other

→ 1:1

→ 1:M

→ M:1

→ M:M

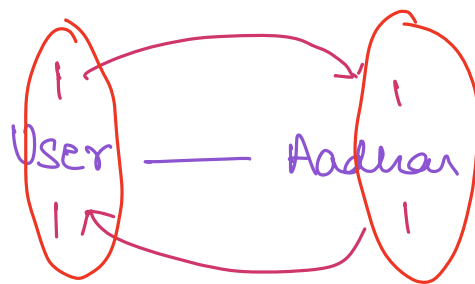
Users

	aadhar_no
---	1234---

OR

aadhar

	user-id
	44



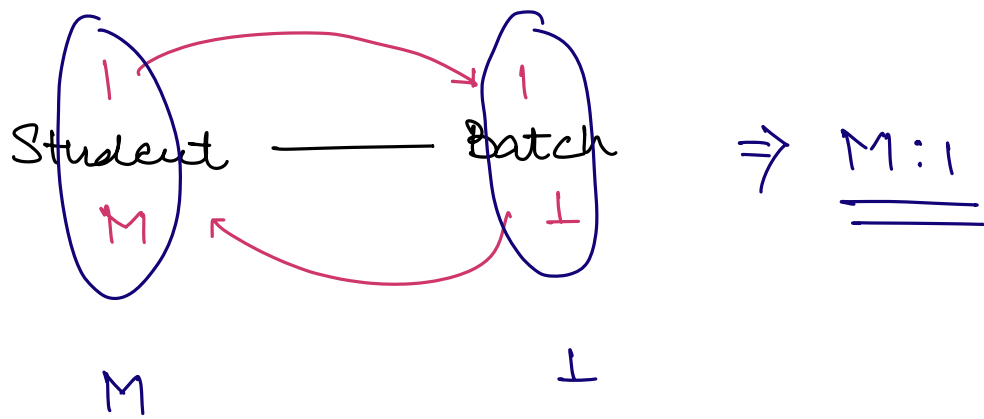
$\Rightarrow 1:1$



A : B

1 : 1

$\Rightarrow$ Id of one side on other side.



Students		OR		batches	
	batch-id				st-ids
107 ABC — —	1024			1024 AIML —	[- , - , - , - , -]



A — B
M : 1

$\Rightarrow$ Id of 1 side on M side.

1:M / M:1 $\nearrow$

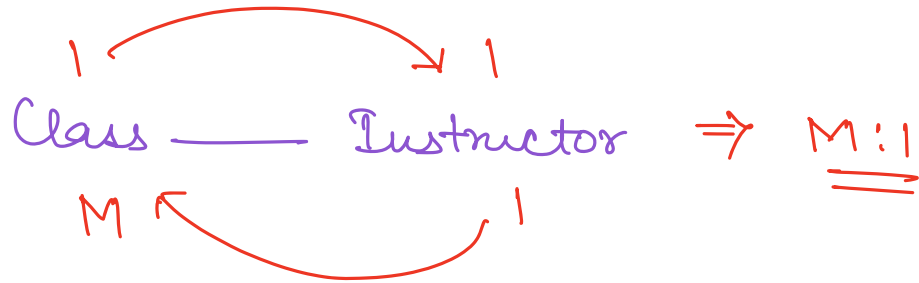
Classes

	ins-id

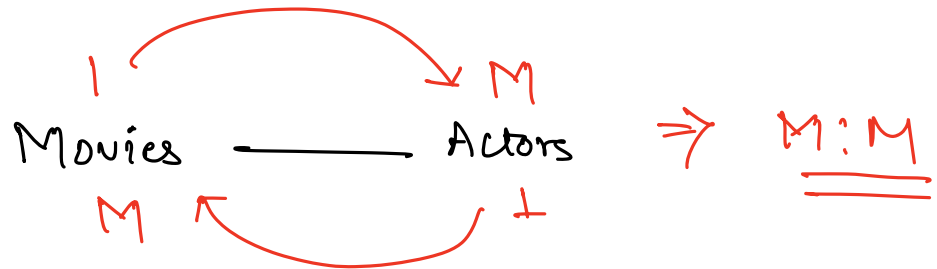
OR

instructors

	list<class>
[-----]	



⇒



movies

[-----]	

OR

actors

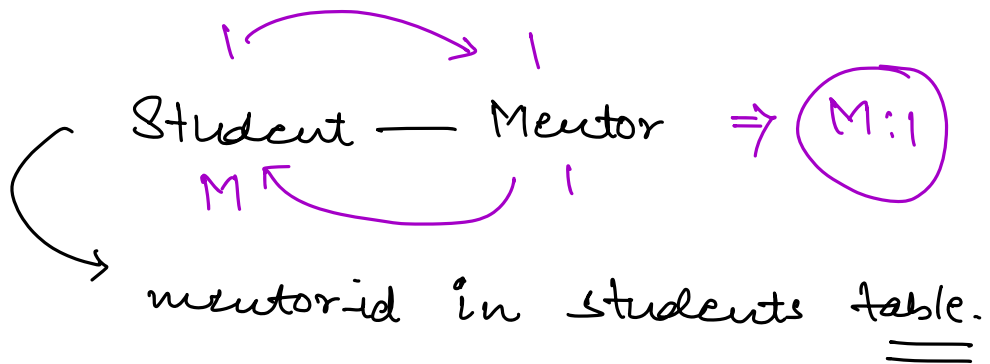
[-----]	

→ Mapping Table

movies_actors

movie_id	actor_id
100	70
100	50
100	47
100	32
101	—
—	—
—	—

M:M ⇒ Mapping Table



batches

					batch-type
—	—	—	—	—	Data Science
					Machine Learning
					Data Science
					Data Science
					Software Dev <u>Eng</u>
					Machine Learning
					Machine Learning
					Machine Learning
					Machine Learning
					Software Dev <u>Eng</u>

} Space Waste

batch-type $\begin{cases} \text{Machine Learning} \\ \text{Data Science} \\ \text{Software Dev } \underline{\text{Eng}} \end{cases}$

Enums. $\rightarrow$ Fixed Set of Values.

batches

				batch-type-id
—	—	—	—	1
				1
				1
				1
				1

batch-type

id	Value
1	Machine Learning
2	Data Science
3	Software Dev _{bug}

